# Desca Affajry Rais

**Mathematics Undergraduate · Researcher · Educator**

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## PROFESSIONAL SUMMARY

I am a final-year Mathematics undergraduate at Universitas Diponegoro with a 3.92/4.00 GPA. My primary interests are pure mathematics, especially real and complex analysis, graph theory, abstract algebra, topology, and combinatorics. I also have experience in university mathematics teaching, olympiad coaching, academic-content development, data analysis, and computational projects.

## EDUCATION

Sep 2023 — Present
ONGOING

### Universitas Diponegoro

**B.Sc. in Mathematics · GPA 3.92 / 4.00**

Chair, UKM KOMIPA UNDIP (2025); teaching and tutorial experience in Real Analysis II; active in the Mathematics Student Association; ON-MIPA delegate and mathematics competition awardee.

Jun 2020 — Mar 2023
COMPLETED

### SMA Negeri 1 Sulang

**Mathematics and Natural Sciences · Final Grade 86.55 / 100.00**

Earned regional and national mathematics awards, represented the school in OSN Informatics, and provided peer mathematics mentoring.

## RESEARCH & ACADEMIC WORKS

2026
DRAFT

### Distance Spectra of Corona Graphs of Complete Graphs and Paths via Invariant Subspaces

**Desca Affajry Rais · Suryoto, S.Si., M.Si.**

Studies the distance spectrum of $K_n \odot P_m$ using block matrices, Kronecker products, and invariant subspaces.

2026
DRAFT

### Connectivity of Int-Graphic Topology on Directed Graphs

**Desca Affajry Rais · YD Sumanto · Solikhin**

Characterizes connectivity of finite int-graphic Alexandroff spaces through minimal open neighborhoods and the out-neighborhood inclusion preorder.

2026
PRESENTED

### Modular Total Edge Irregular Labeling of Corona Product of Cycle $C_4$ with Path and Null Graphs

**The 16th ISNPINSA · Oral Presentation**

Developed explicit labeling constructions and established exact modular total edge irregularity strength results.

2026
COMPLETED

### Inequality Problem Solving Book

**Desca Affajry Rais · Drs. YD Sumanto, M.Si. · First Edition · 174 pages**

Proof-oriented mathematics book covering inequality theory, rigorous proofs, worked examples, and exercises. The manuscript is complete; Indonesian Intellectual Property (HKI) registration is planned.

## PROFESSIONAL EXPERIENCE

Aug 2026 — Present
ONGOING

### University Mathematics Tutor

**Tutorkuliah.id**

Tutoring in advanced undergraduate mathematics with emphasis on conceptual understanding and rigorous problem solving.

Jan 2026 — Present
ONGOING

### Mathematics Olympiad Coach

**KOMIPA UNDIP**

University-level training in Real Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, and Combinatorics.

Apr 2026 — Jun 2026
COMPLETED

### Teaching Assistant

**Asosiasi Matematika Terapan UNDIP**

Real Analysis tutorials, proof-oriented problem solving, and academic consultation.

Dec 2025 — Mar 2026
COMPLETED

### Mathematics Item Writer

**SainsIn**

Authored quantitative-reasoning items, detailed solutions, instructional videos, and quality-control materials.

Nov 2025 — Jan 2026
COMPLETED

### Data Analyst
**Emergencyy Call**
Data cleaning, SQL querying, Excel reporting, KPI dashboards, visualization, and workflow documentation.

Nov 2025 — Dec 2025
COMPLETED

### Mathematics Item Writer
**SNBT.ID**
Authored high-difficulty Quantitative Knowledge and Mathematical Reasoning items with structured solutions.

May 2025
COMPLETED

### Project-Based Virtual Intern · Data Scientist
**ID/X Partners × Rakamin Academy**
Built a credit-risk prediction pipeline using Logistic Regression and Random Forest.

Mar 2025
COMPLETED

### Project-Based Virtual Intern · Big Data Analyst
**PT Kimia Farma Tbk × Rakamin Academy**
Built a BigQuery analytical table and a Looker Studio performance dashboard.

## LEADERSHIP & ORGANIZATIONS

Jan 2025 — Dec 2025
COMPLETED

### Chairman
**Mathematics and Sciences Olympiad Club (KOMIPA UNDIP)**
Led strategy, daily operations, olympiad mentoring, ON-MIPA delegation, and external representation.

Jan 2025 — Dec 2025
COMPLETED

### Senior Staff, Education and Reasoning Department
**Mathematics Student Association, Universitas Diponegoro**
Delivered mathematics tutorials, educational content, and learning-material development.

Feb 2024 — Jan 2025
COMPLETED

### Junior Staff, Education and Reasoning Department
**Mathematics Student Association, Universitas Diponegoro**
Coordinated Academic Delegation, mentoring for 100+ students, and received Best Junior Staff and Best Department Staff recognition.

Oct 2023 — Dec 2024
COMPLETED

### Staff, Networking and Public Relations Department
**KOMIPA UNDIP**
Supported ON-MIPA selection, served as Mathematics Liaison Officer, and led UKM Roadshow 2024.

2020 — 2023
COMPLETED

### Moderator, Mentor, and Board of Governor
**Brainly**
Progressed through multiple moderation and mentoring roles, including supervision of 29 apprentice moderators.

2020 — 2022
COMPLETED

### Organizational Leadership
**Pelajar Islam Indonesia (PII), Rembang**
Served as Head of Organizational Development and later General Secretary of the Rembang Regional Board.

## COMMITTEE & VOLUNTEER EXPERIENCE

**Facilitator Division Coordinator / Koordinator Divisi Fasilitator**
27th Undip's Mathematics Competition · Jan 2025 — Feb 2026

**Curriculum Division Staff / Staf Divisi Kurikulum**
LKMM-TD FSM 2025 · Jan — Oct 2025

**Event Division Staff / Staf Divisi Acara**
Introduction, Gathering, & Adaptation Mathematics 2025 · Mar — Oct 2025

**Event Division Staff / Staf Divisi Acara**
Industrial Visit Mathematics 2025 · Mar — Aug 2025

**Equipment Division Staff / Staf Divisi Perlengkapan**
Math Action for Social and Education 2025 · Mar — Jun 2025

**Public Relations Division Staff / Staf Divisi Humas**
Leadership Training FSM 2025 · Mar — Apr 2025

**Event Division Coordinator / Koordinator Divisi Acara**
26th Undip's Mathematics Competition · Mar 2024 — Feb 2025

**Chief Executive / Ketua Pelaksana**
UKM Roadshow 2024 · Jun — Sep 2024

**Volunteer Teacher / Pengajar Sukarela**
Mathematics Clinic 2024 · Aug 2024

**Equipment Division Coordinator / Koordinator Divisi Perlengkapan**
ON-MIPA UNDIP Selection 2024 · Oct 2023 — Feb 2024

**Mathematics Liaison Officer / Liaison Officer Matematika**
ON-MIPA Coaching Center 2024 · Feb 2024

## SELECTED PROJECTS

Feb 2026
COMPLETED

### Crowd Optimization and Smart Routing Strategy at Dufan Ancol
**Operations Research · Mathematical Modeling**

Applied M/M/s queueing, Little's Law, greedy optimization, and knapsack concepts; proposed 14,000 daily visitors, a <45-minute waiting-time target, and 75% maximum utilization.

Feb 2026
ONGOING

### DMath Learning
**Mathematics Education · EdTech**

Ongoing mathematics learning platform for structured materials, guided practice, problem banks, olympiad preparation, and learning tools.

Feb 2026
COMPLETED

### Bike Sharing Analytics
**Python · EDA · Streamlit**

Identified weekday demand peaks, weekend casual-user behavior, and environmental drivers through end-to-end exploratory analysis.

Dec 2025
COMPLETED

### Financial Anomaly Detection System
**Scikit-Learn · K-Means · Random Forest**

Combined unsupervised pattern discovery with supervised transaction classification.

May 2025
COMPLETED

### Credit Risk Prediction Analysis
**SMOTE · Logistic Regression · Random Forest**

Analyzed 466,285 observations and reported Random Forest performance of 99.70% accuracy, 99.94% precision, and 99.46% recall.

Mar 2025
COMPLETED

### Kimia Farma Business Performance Analysis
**BigQuery · SQL · Looker Studio**

Built a centralized analytical table and multidimensional performance dashboard.

Mar–Apr 2025
COMPLETED

### Investment Portfolio Optimization
**Markowitz · Efficient Frontier · Sharpe Ratio**

Analyzed portfolio allocation across KLBF.JK, HEAL.JK, and KAEF.JK using risk–return optimization.

## SCHOLARSHIPS & TRAINING

Nov 2024 — Present
ONGOING

### BSI Scholarship Inspirasi Batch 4
**Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.**

Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Feb — Jul 2026
COMPLETED

### Coding Camp 2026 — Data Scientist
**Coding Camp powered by DBS Foundation**

Coding Camp powered by DBS Foundation

Mar — Jun 2026
COMPLETED

### Microsoft Elevate Training Center
**Data Science with Microsoft Fabric · Generative AI with Microsoft Azure**

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Jan — Mar 2026
COMPLETED

### IDCamp 2025 — Intermediate Data Scientist
**Indosat Ooredoo Hutchison**

Indosat Ooredoo Hutchison

Oct — Dec 2025
COMPLETED

### IDCamp 2025 — Beginner Data Scientist
**Indosat Ooredoo Hutchison**

Indosat Ooredoo Hutchison

## CERTIFICATIONS

**UNDIP SEU English Proficiency Test**
EPT 65 · TOEFL Prediction equivalent 549 · 17 Sep 2026

**TOEFL ITP Prediction Test**
590 / 677

**AI Engineer Bootcamp / Bootcamp AI Engineer**

**Foundations of Artificial Intelligence / Belajar Dasar AI**

**Foundations of SQL / Belajar Dasar SQL**

**Computer Networking Fundamentals / Fundamental Jaringan Komputer**

**Data Analyst — Basic Level**
ITBOX · Mar 2025

**Python Programming Essentials: A Beginner's Guide**
Rakamin Academy · Mar 2025

**Classical Cryptography for Beginners**
Cyber Academy Indonesia · Jul 2024

**Introduction to Information Security**
Cyber Academy Indonesia · Jul 2024

**Student Leadership and Management Training — FSM 2024**
BEM FSM UNDIP · Apr 2024

**Introduction to Data Science with Python**
DQLab · Jan 2024

**Python Fundamentals for Data Science**
DQLab · Jan 2024

**Basic Student Management Skills Training / LKMM Pra Dasar**
Sep 2023

**Python Programming for Beginners**
CODEPOLITAN · Credential Z7R8TDD · Jul 2024 — Jul 2027

## SKILLS

**Mathematics & Theory**

Pure Mathematics · Real Analysis · Complex Analysis · Graph Theory · Abstract Algebra · Topology · Combinatorics · Mathematical Modeling · Formal Proof Writing

**Technical & Computing**

Python · SQL · LaTeX · Scikit-Learn · Looker Studio · Microsoft Excel · Microsoft Fabric · Microsoft Azure

**Data & Analytics**

Data Analysis · Exploratory Data Analysis · Machine Learning · Data Visualization · Statistical Analysis · Data Engineering

**Academic & Teaching**

Research · Academic Writing · Mathematics Olympiad · Teaching · Curriculum Development · Assessment Design · Problem Solving

## AWARDS & ACHIEVEMENTS

**2nd Place / Juara 2 · Calculus Competition · Universitas Halu Oleo**
Oct 2026

**Silver Medalist / Medali Perak · Isaac Newton Competition 2026 · OLIMNUS × UIN Sunan Kalijaga**
Jul 2026

**Silver Medalist / Medali Perak · Future Science Olympiad · POSI**
Jun 2026

**Bronze Medalist / Medali Perunggu · Mathematical Analysis and Geometry Day XVII · Institut Teknologi Bandung**
May 2026

**Gold Medalist / Medali Emas · Cendekia Natural Science Competition**
Apr 2026

**1st Place / Juara 1 · ON-MIPA UNDIP Selection 2026**
Apr 2026

**1st Place / Juara 1 · Indonesia Science Competition (KSI)**
Mar 2026

**Gold Award · KTOM March 2026**
Mar 2026

**1st Place / Juara 1 · Indonesia Science Competition (InSC)**
Feb 2026

**4th Place · Honorable Mention 1 · MATICS Mathematics Olympiad · Hasanuddin University**
Feb 2026

**Silver Award · KTOM February 2026**
Feb 2026

**Bronze Medalist / Medali Perunggu · POSI Science Competition**
Jan 2026

**2nd Place / Juara 2 · Indonesian Youth Science Olympiad #6**
Jan 2026

**2nd Place / Juara 2 · National Science League #2**
Jan 2026

**Silver Medalist / Medali Perak · Einstein Science Competition 2026**
Jan 2026

**Gold Award · KTOM December 2025**
Dec 2025

**Special Honour Participant · International Youth Math Challenge**
Dec 2025

**1st Place / Juara 1 · LOSEMA X**
Nov 2025

**1st Place / Juara 1 · Olimnus Science 2025**
Nov 2025

**Silver Medalist / Medali Perak · Isaac Newton Competition 2025**
Jul 2025

**3rd Place / Juara 3 · ASOE 4.0**
May 2025

**4th Place · ON-MIPA UNDIP Selection 2025**
Mar 2025

**Finalist / Finalis · MATHLICIOUS 2022**
Nov 2022

**1st Place / Juara 1 · ALHAMBRA Mathematics Competition**
Feb 2022

**Bronze Medalist / Medali Perunggu · Insight Arjuna Competition**
Oct 2021

**Gold Medalist / Medali Emas · Dirgantara Science Competition**
Oct 2021

**Silver Medalist / Medali Perak · Indonesian Youth Science Competition**
Oct 2021

**Bronze Medalist / Medali Perunggu · Bimage Competition Season 2**
Sep 2021

**Silver Medalist / Medali Perak · POSI Science Competition**
Sep 2021

**Top 10 · National Calculus Olympiad Nusantara**
Aug 2021